User Manual

This is a simple debit credit system for a banks savings account. End users at different branches of the bank maintains the customers and their accounts. Admins maintains the end users.

The features that this system contains:

- Two panels: User & Admin
- Login Logout Session
- Admin can create user account User gets the password through email
    - Password is generated randomly with alphanumeric value
    - User is created after checking the employee list
- Admin can reset the user password
- Admin can assign role to user
- User can create customer account
- User can change their password
- User can deposit/withdraw money
- User can search customer by Account No
- Form validations
    - Empty Field Validation
    - Email Format Validation
    - Minimum Inserted Balance should be 500 while creating account
    - Browsers Back button shouldn't work after logging out
    - Error message for wrong user id or password

The features that will be added later:

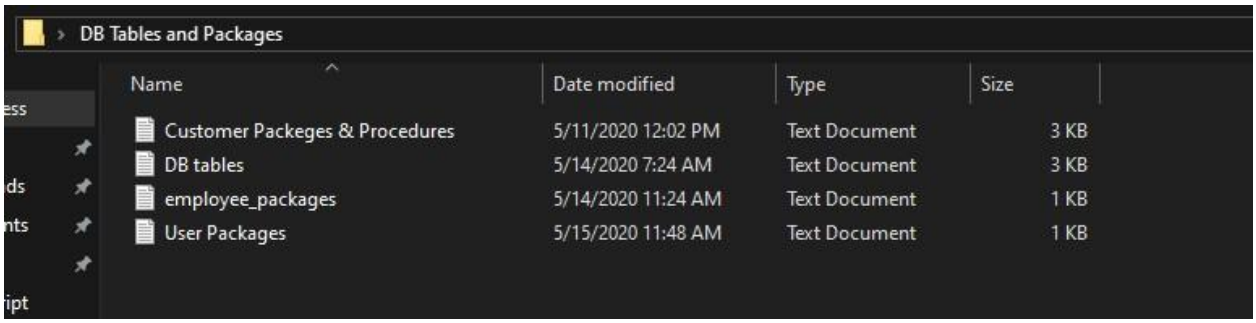
- Admin will be able to create menu
- Admin will be able to create role
- Admin will be able to assign menu to role
- User will generate customer list report
- User will be able to upload photos and NID of customers holding savings account

IF YOU ARE ABSOLURELY NEW WITH ORACLE:

1. Go to SQL command Line
2. Sql > connect;
3. Provide User name & Password
4. Then you start giving the table and package command and continue according to the user manual bellow.

How to Run the system

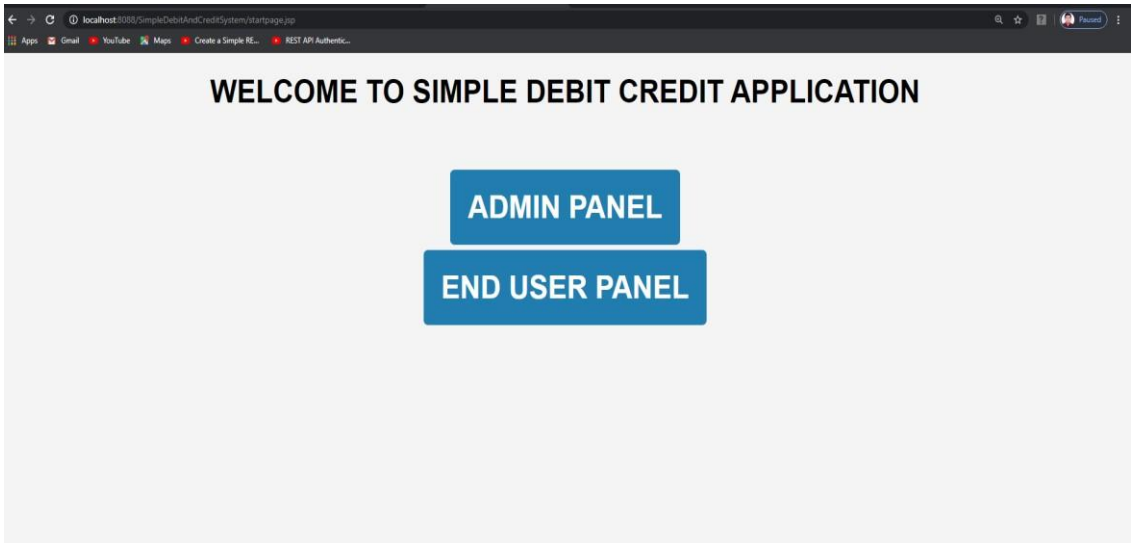
1. First set up the necessary environment and create the tables and packages attached in the text file.



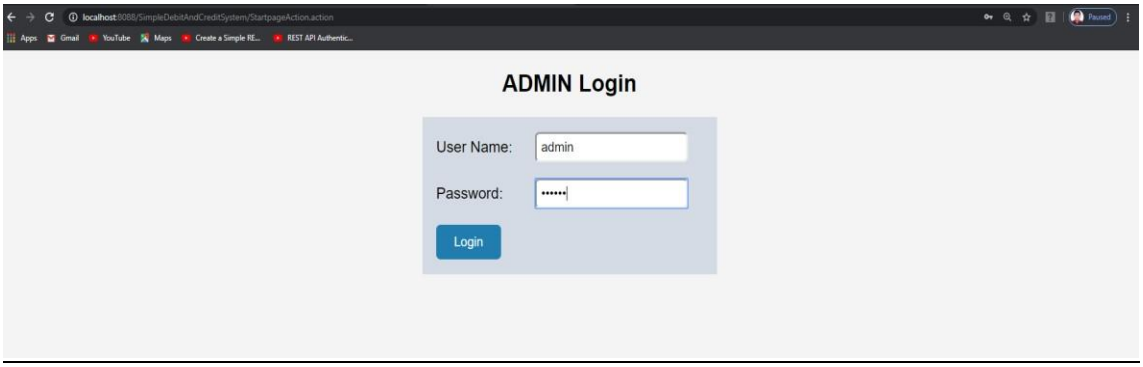
2. Run the "startPage.jsp" on the server. Or simply run the tomcat server and type the url bellow in the search bar of your web browser.

   "http://localhost:8088/SimpleDebitAndCreditSystem/startpage.jsp"

   Output:

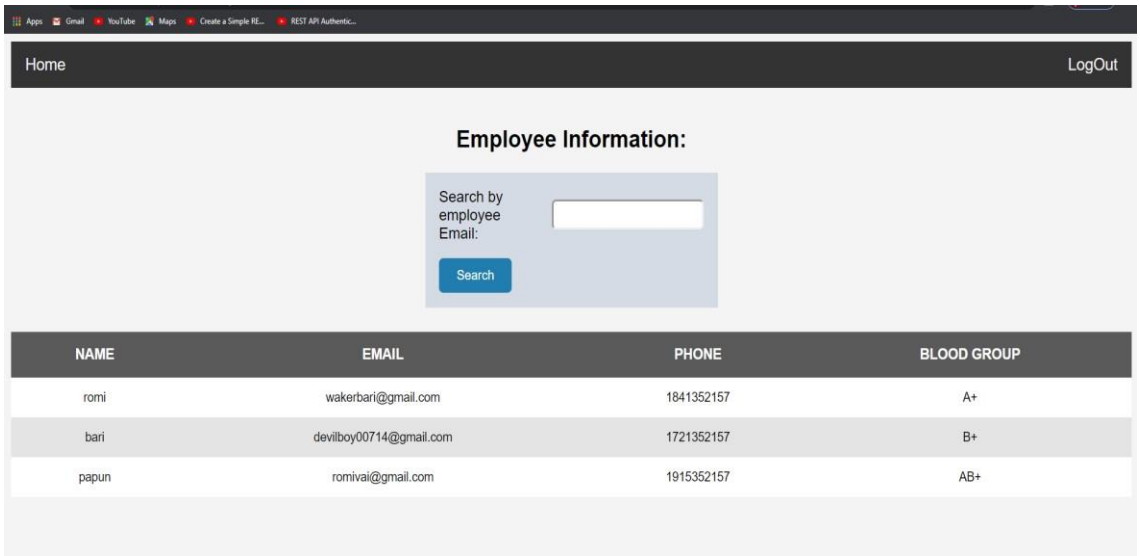


3. Lets select Admin Panel from point 2. An Admin login form will arrive.
   Output:



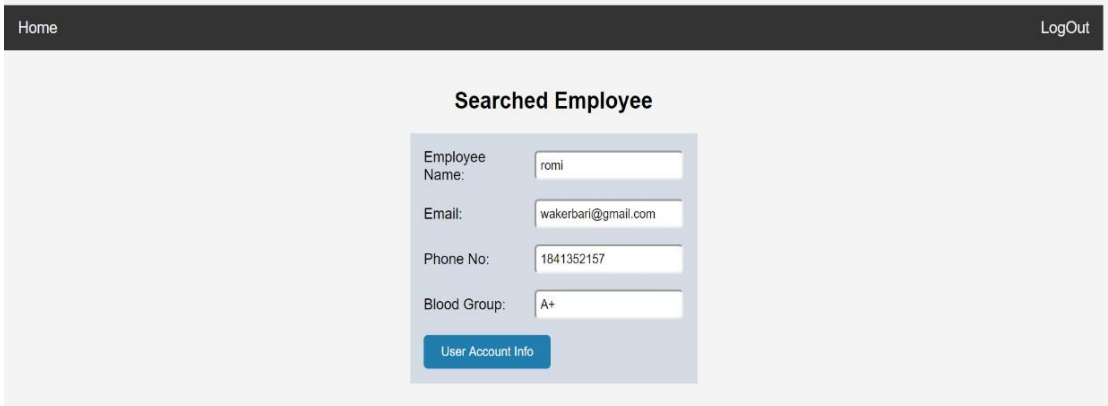
4. After getting in with appropriate user id and password we'll find a jsp page where the list of the employees will be shown. And there will be a search bar above the table where we can find the employee with their email id to verify if they have any user id for logging in the system or not.
   If they do the admin has the option to change their password.
   Else if they don't the admin has the option to create an user account for them.

   Output:

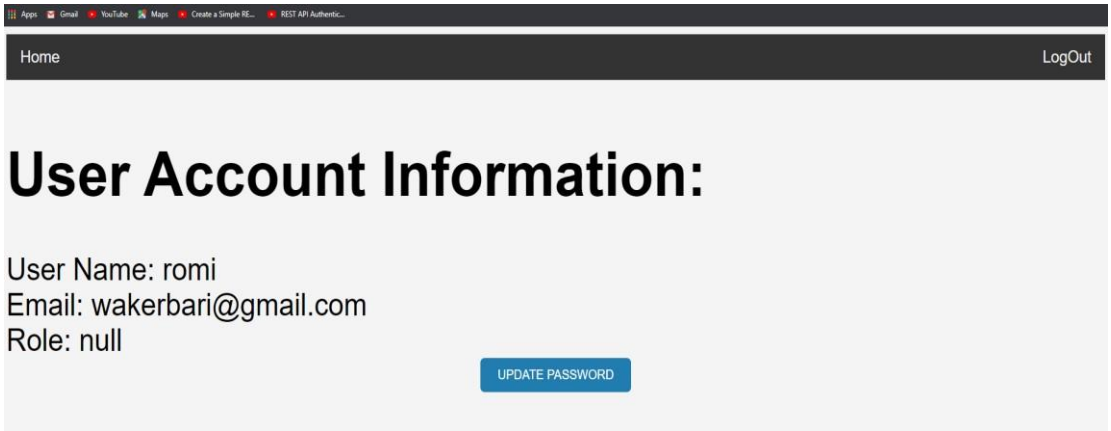


→First lets search with "wakerbari@gmail.com". An end user id exits with this email id.

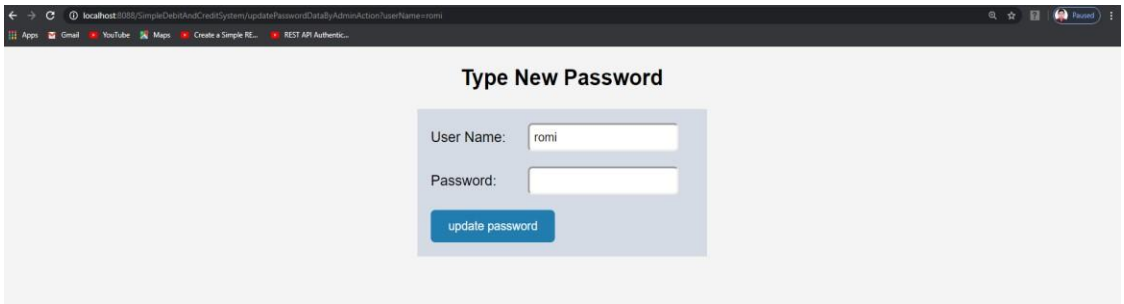
Output:



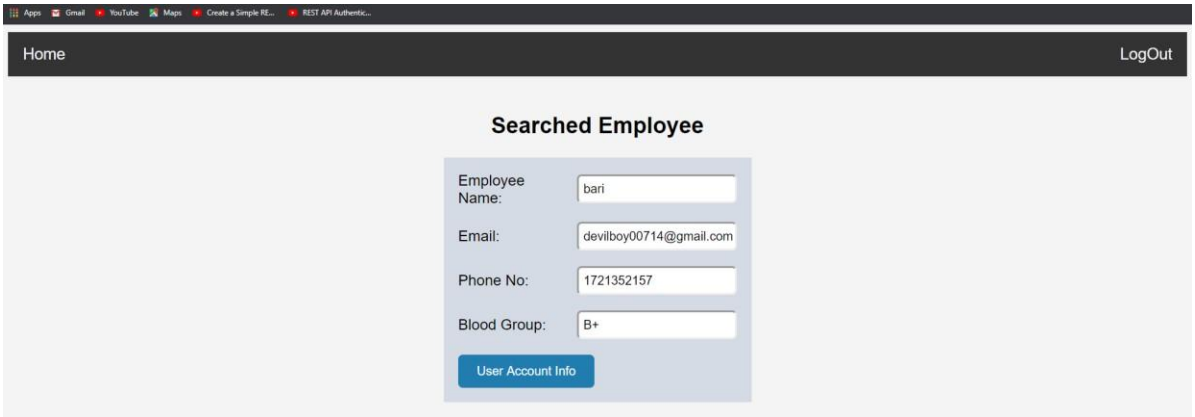
Press "User Account Info" Output:



Press "Update Password" Output:

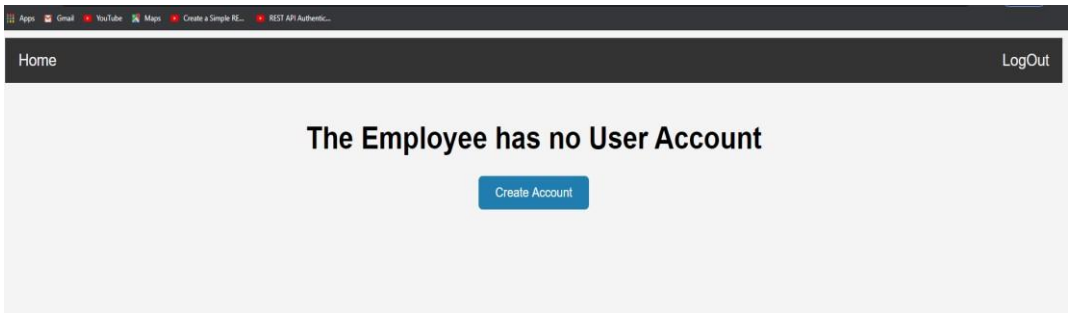


→Now lets search with the id "devilboy00714@gmail.com". There is no end user id with this email.



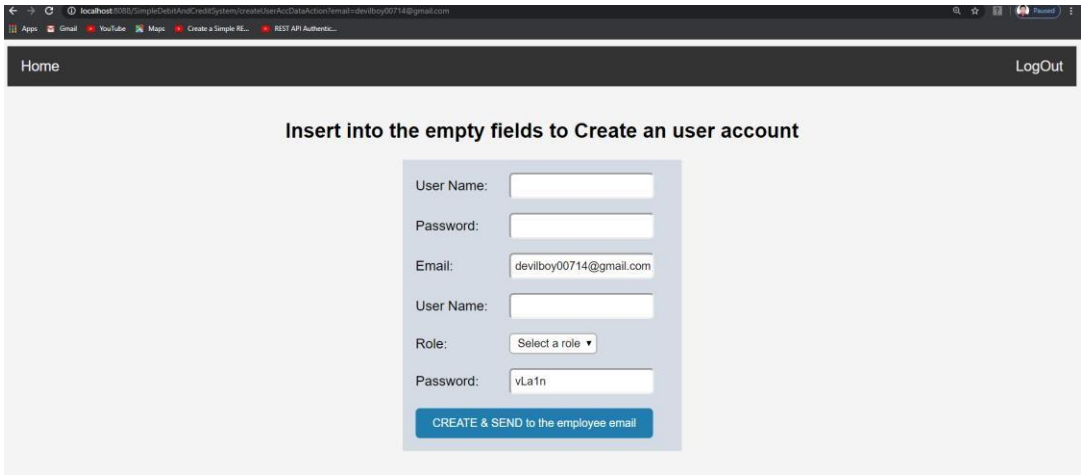
Press “User Account Info”

Output:



As there is no existing user account, admin will be given option to create one.

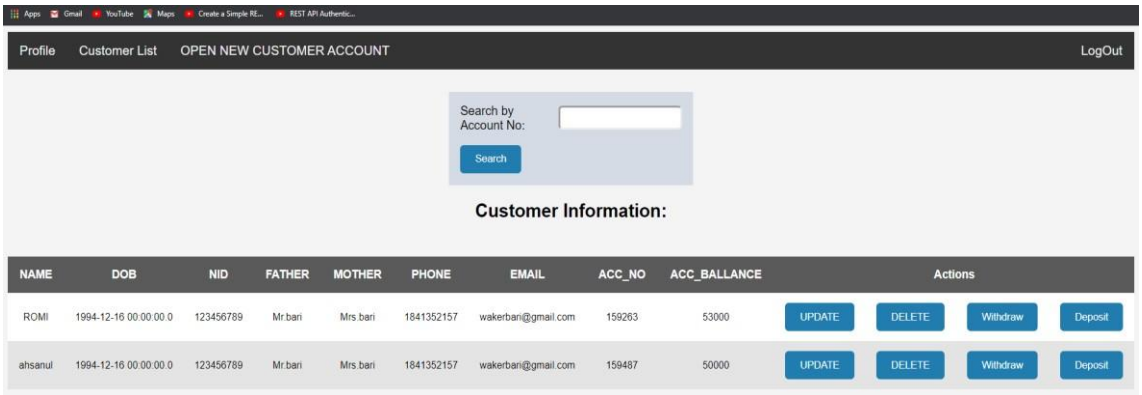
Output:



As we can see a random password is generated and the employee email where the informations(user id and password) will be sent after filling up the fields and pressing “Create and send”.

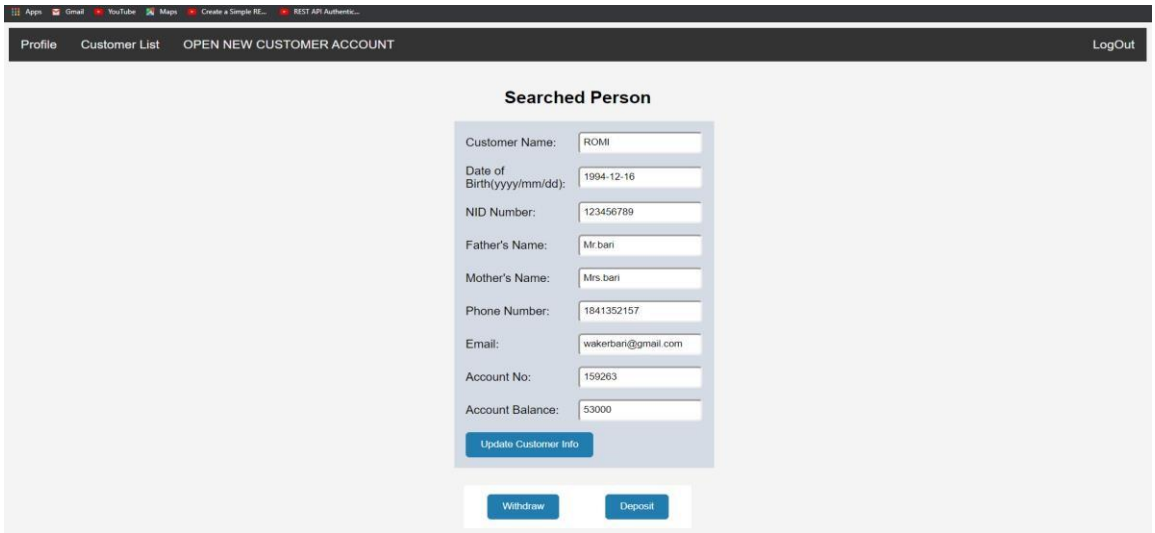
5. Now from point 2, if we had chosen “End User Panel”. A user login form will arrive as expected.
   After providing proper user id and password we will get into the user panel and at home page we’ll find a “list of Customers”.

   Output:



6. The is “Update” option to update email or phone number or some other information that can be updated. We must understand that the account number shouldn’t be allowed to be updated.
   And the other necessary options “Delete” and customer account. Withdraw or Deposit money to the customer account.

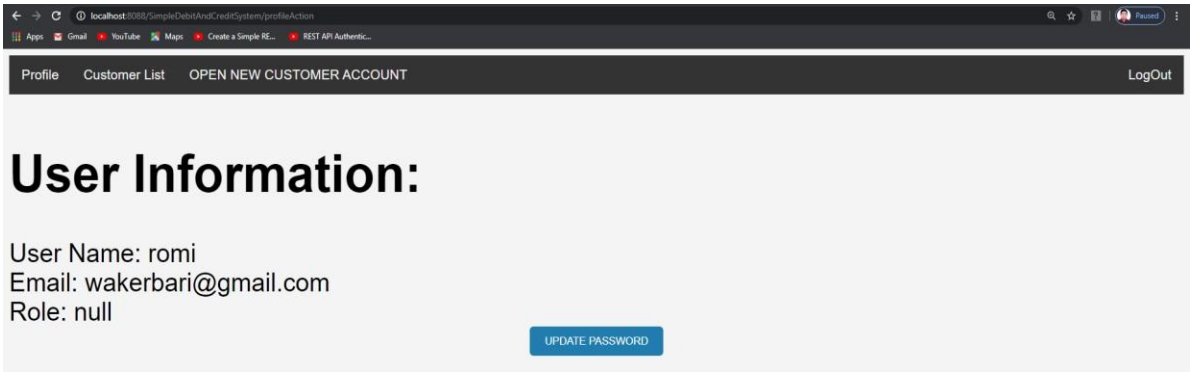
7. Now there is a search option on top where we can search using the “Account Number”.
   Output:

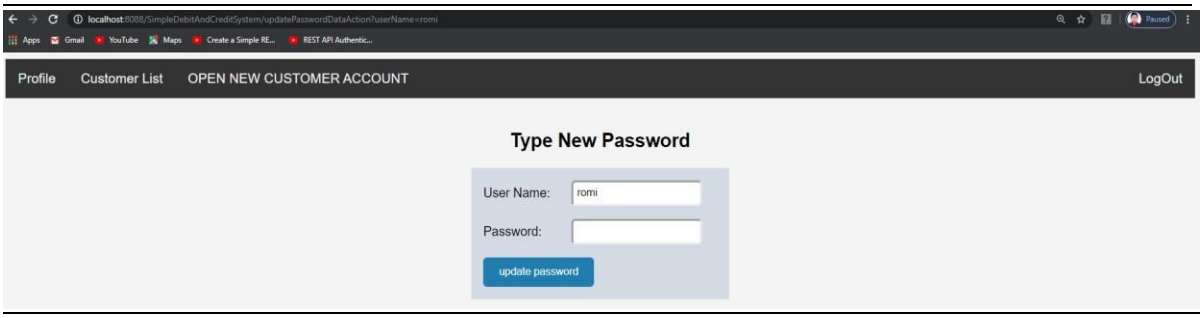


   So after searching out the customer the end user has the option to Update info, withdraw and deposit.

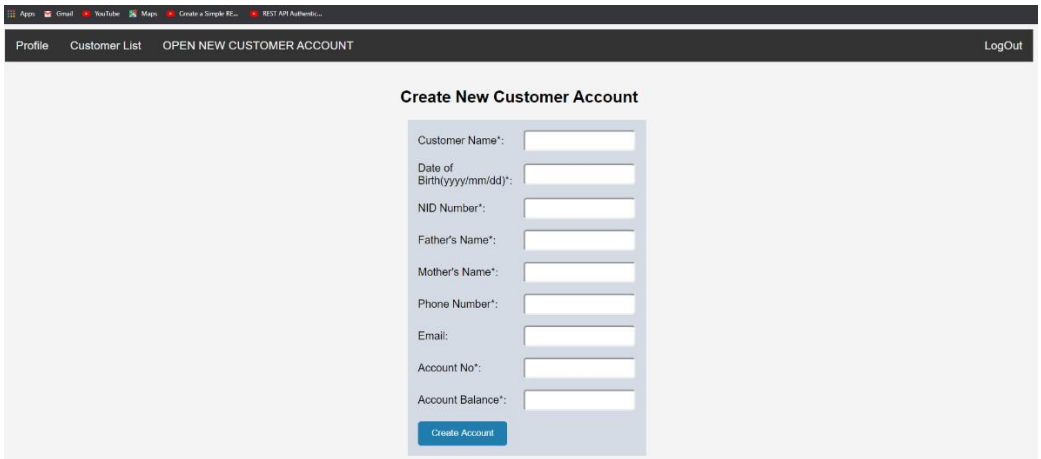
8. Now if we notice the Navigation bar there are some important features here. “Profile” takes us to the user profile who is currently logged into the system.
   And there the user will find the option to change the auto generated password of their User id.

   Output:





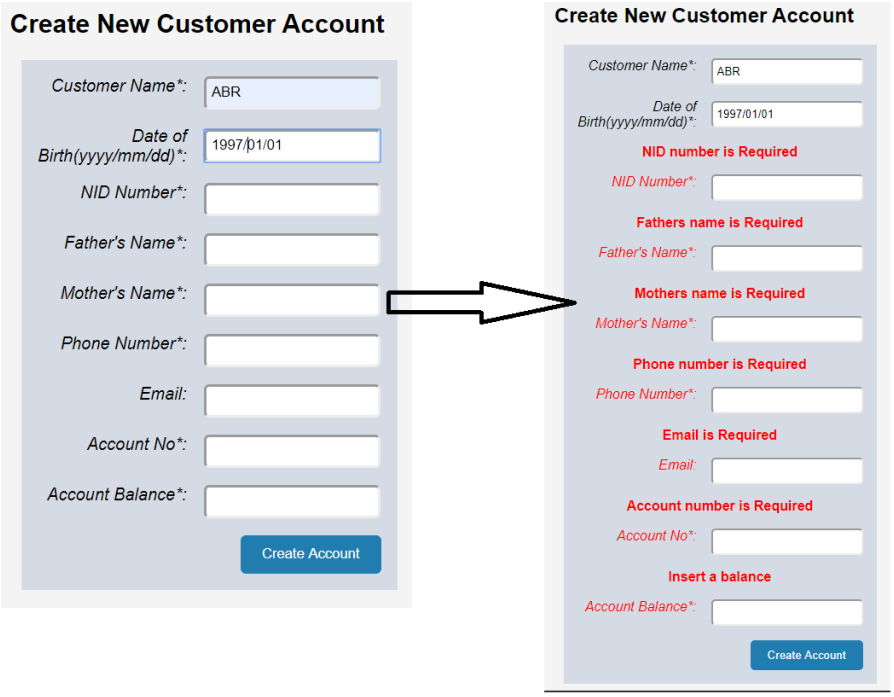
9. Another option we can see in the navigation bar is “Create New User”. It allows the end user to create a new account for a customer. Output:



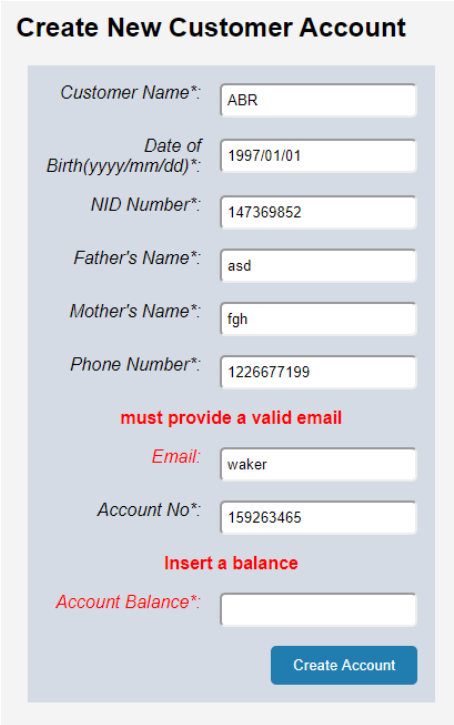
10. Both the panel holds the “Logout” mechanism.
    Session is created with Java Servlet API

VALIDATIONS:

➔ Empty field validation :



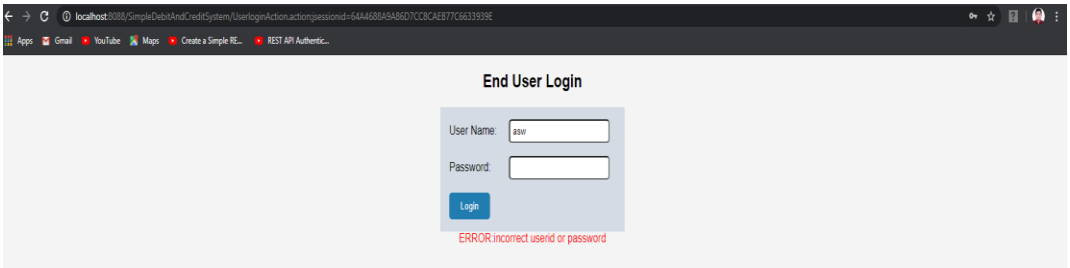
➔ Valid Email Format :



➔ Minimum Balance Required

➔ Browsers back button won't work after logging out

- Disabling the BACK BUTTON could be a solution
- But the best way is to control the cache and revalidate every time before loading

```
struts.xml   addCustomer.jsp   AddCustomerAction.java   AddCustomerAction-validation.xml   userLogin.jsp   UserLogoutAction.java   header.jsp
 2      pageEncoding="ISO-8859-1"%>
 3
 4      <%@ taglib prefix="s" uri="/struts-tags" %>
 5
 6  <!DOCTYPE html>
 7  <html>
 8
 9  <head>
10  <meta charset="ISO-8859-1">
11  <title>header</title>
12  </head>
13  <body>
14  <%
15  response.setHeader("Cache-Control", "no-cache, no-store, must-revalidate"); //HTTP 1.1
17  response.setHeader("Pragma", "no-cache"); //HTTP 1.0
18  response.setHeader("Expires", "0"); // Proxies
19
20
21  if(session.getAttribute("loggedinUser")==null){
22      response.sendRedirect("userLogin.jsp");
23  }
24
25  %>
26
27      <div class="topnav">
28        <a href="profileAction">Profile</a>
29        <a href="userWelcomeAction">Customer List</a>
30        <a href="addCustomer.jsp">OPEN NEW CUSTOMER ACCOUNT</a>
31        <a href="userLogoutAction" style="float:right">LogOut</a>
32      </div>
33  </body>
34  </html>
```

➔ Error message for wrong user id or Password